

MA341

APPLIED ODE'S

Handout 2

Definition 1. Bernoulli Equation is a non linear equation of the form $y' + a(x)y = b(x)y^m$, where $m \neq 0, 1$. Notice that $y = 0$ solves the equation in case of $m > 0$, but not when $m < 0$. We are to find a non trivial (non zero) solution to Bernoulli equation. Rewrite the equation as $y'y^{-m} + a(x)y^{1-m} = b(x)$.

Define $z = y^{1-m}$, differentiate $z' = (1-m)y^{-m}y'$ and substitute into equation:

$$\frac{z'}{1-m} + a(x)z = b(x).$$

This is a linear equation, which standard form reads for

$$z' + (1-m)a(x)z = (1-m)b(x),$$

which solved with the help of integration factor $\mu = e^{(1-m)\int a(x)dx}$ to get

$$y^{(1-m)} = z = (1-m)e^{-(1-m)\int a(x)dx} \int e^{(1-m)\int a(x)dx} b(x) dx + C.$$

Definition 2. Riccati Equation is another non linear equation, its form is $y' = b(x)y^2 + a(x)y + c(x)$. For $c = 0$ this is a private case of Bernoulli equation, therefore we assume $c \neq 0$. Suppose that we know that $y_1(x)$ is a known private solution. To find a general solution define $y = y_1 + u$ and differentiate to get Bernoulli equation:

$$\begin{aligned} y' &= y_1' + u' = b(x)(y_1 + u)^2 + a(x)(y_1 + u) + c(x) \\ \Rightarrow u' &= b(x)(y_1^2 + 2y_1u + u^2) + a(x)(y_1 + u) + c(x) - (b(x)y_1^2 + a(x)y_1 + c(x)) \\ \Rightarrow u' &= b(x)(2y_1u + u^2) + a(x)u \\ \Rightarrow u' - (b(x)2y_1 + a(x))u &= b(x)u^2 \end{aligned}$$

Alternatively one uses $y = y_1(x) + \frac{1}{u(x)}$ to get directly into linear ODE(verify!):

$$\begin{aligned} -\frac{u'}{u^2} &= b(x)\left(2\frac{y_1(x)}{u(x)} + \frac{1}{u^2(x)}\right) + a(x)\frac{1}{u(x)} \\ u' + (2b(x)y_1(x) + a(x))u &= -b(x) \end{aligned}$$